Institute of Technology of Cambodia
Department of Information and Communication Engineering

Lecture 01

Introduction to Network Security

Network Security

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Content

This lesson is organized into a 60-minute body of three parts, framed by a warm-up, a case study, and a check for understanding.

1

Part 01 · 20 minutes

What is Network Security? *Definitions, scope, and the security mindset*

2

Part 02 · 20 minutes

The CIA Triad *Confidentiality, Integrity, Availability — and beyond*

3

Part 03 · 20 minutes

Common Network Threats *Malware, social engineering, DoS, and insiders*

Learning Outcomes

By the end of this lesson, students will be able to:

1. Define network security and explain its importance in modern systems.
2. Describe the CIA triad (Confidentiality, Integrity, Availability).
3. Identify common types of network security attacks.
4. Relate real-world security breaches to theoretical concepts.

Keywords

Network Security
The policies, practices and technologies that protect a network and the data travelling across it from unauthorized access, misuse, modification or disruption.

CIA Triad
The three core goals of security: Confidentiality, Integrity and Availability. Most security controls exist to protect one or more of them.

Confidentiality
Keeping information secret from anyone who is not authorized to see it, for example by using encryption and access control.

Integrity
Making sure data is accurate and has not been changed, whether by accident or by an attacker, while it is stored or sent.

Availability
Making sure systems and data are accessible to authorized users when they need them, even during failures or attacks such as DoS.

Threat
Any potential cause of harm to a system, such as a hacker, malware, an insider or a natural disaster.

Vulnerability
A weakness in a system, its design or its configuration that a threat could exploit, such as an unpatched bug or a weak password.

Attack
A deliberate action that exploits a vulnerability to break confidentiality, integrity or availability.

Lesson Plan

The Lesson Body runs a full 60 minutes — three parts of 20 minutes each.

20

minutes

Part 1

Part 1: What is Network Security?

20

minutes

Part 2

Part 2: The CIA Triad

20

minutes

Part 3

Part 3: Common Network Threats

Pre-Quiz / Warm-Up

Discuss each question before revealing the answer.

1. Network security mainly aims to:

a) Install more software
b) Share Wi-Fi passwords
c) Speed up the Internet
d) Protect networks and data from unauthorized access

2. The CIA triad stands for:

a) Confidentiality, Integrity, Availability
b) Control, Integration, Access
c) Communication, Internet, Authentication
d) Cipher, Identity, Access

3. A ransomware attack that blocks access to hospital records violates:

a) Availability
b) Confidentiality
c) Integrity
d) None

4. Phishing is an example of:

a) A hashing algorithm
b) A social-engineering attack
c) A firewall type
d) An encryption method

5. An insider threat comes from:

a) People with authorized access
b) Old cables
c) Power outages
d) External hackers only

Pre-Quiz — Answers

Correct answers are highlighted.

1. Network security mainly aims to:

a) Install more software
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5. An insider threat comes from:

a) People with authorized access ✓
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PART

01

What is Network Security?

20 minutes

IN THIS PART

01 What is Network Security?
02 Why Network Security Matters
03 A Definition of Computer Security
04 Assets, Threats, Vulnerabilities & Controls
05 Information vs. Network Security
06 Security Services & Mechanisms
07 Who Are the Attackers?
08 Thinking About Risk



What is Network Security?

Network security is the practice of protecting computer networks and the data that moves across them from unauthorized access, misuse, or disruption. It grew out of the wider field of information security as computing became connected.



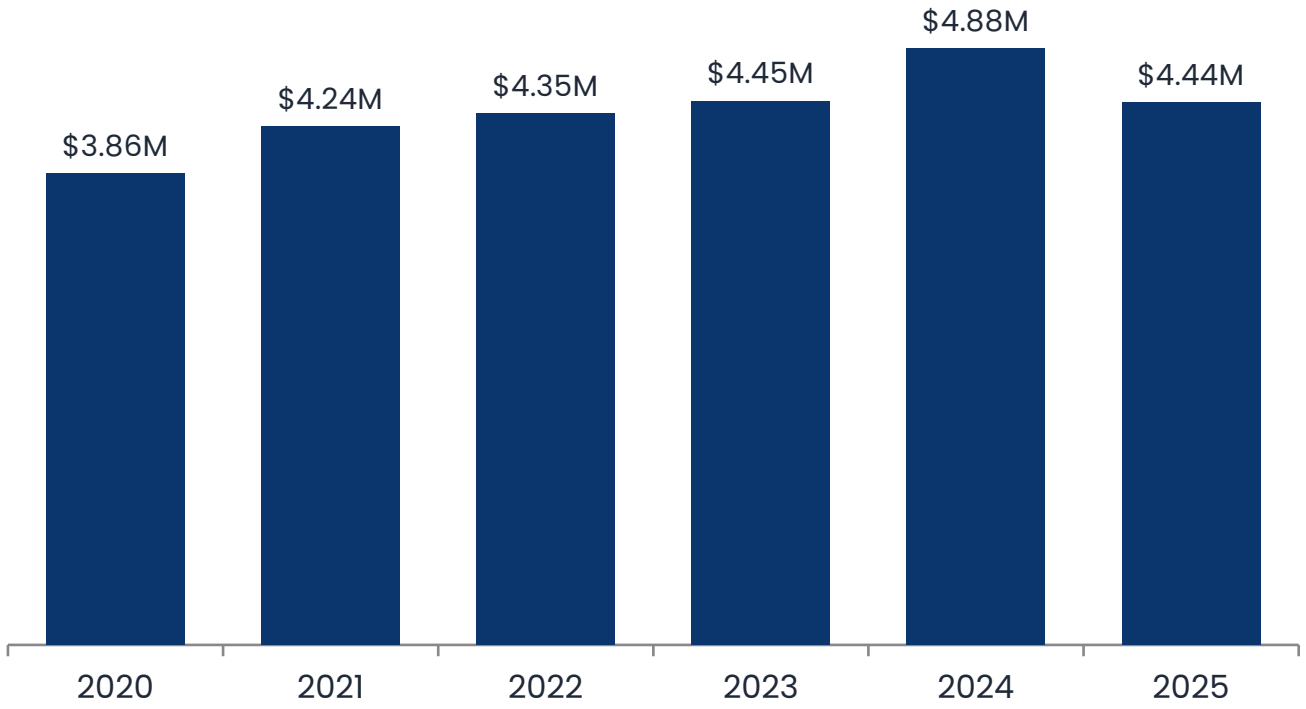
The field evolved

- Information security — physical & administrative controls
- Computer security — automated tools for shared systems
- Network security — protecting data in transit
- Internet security — protection across public networks

Why Network Security Matters

Networks carry the information and services that organizations and individuals depend on. When they fail, the consequences are financial, legal, and personal.

Average cost of a data breach, worldwide (USD millions)



Financial
Losses, recovery costs, fines, lost business.

Trust
Customers and partners lose confidence.

Continuity
Downtime halts operations and services.

Legal
Data-protection laws impose duties and penalties.

Source: *IBM Security, Cost of a Data Breach Report (2020–2025 editions)*

A Definition of Computer Security

The NIST Computer Security Handbook frames computer security around protecting the resources of an information system:

> **Computer security** — measures that preserve the **integrity, availability, and confidentiality** of information-system resources, including hardware, software, firmware, data, and telecommunications.

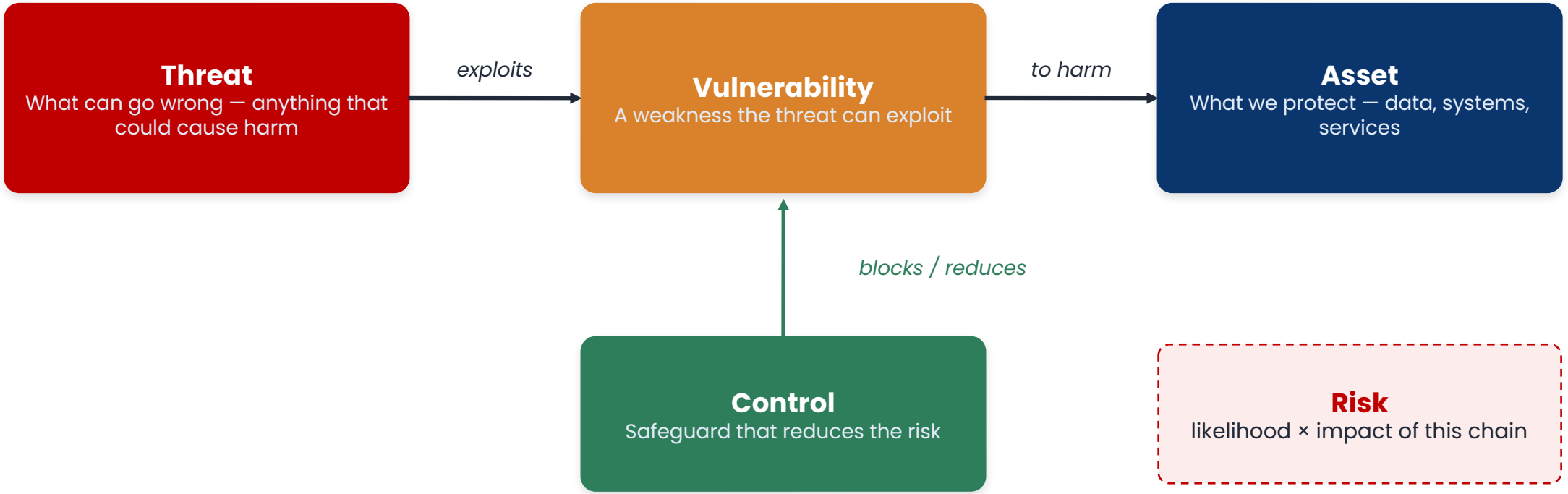
Three core objectives

- Confidentiality: keep data private
- Integrity: keep data correct
- Availability: keep systems usable



Assets, Threats, Vulnerabilities & Controls

Four terms frame every security discussion. The diagram shows how they relate: risk exists when a threat can exploit a vulnerability to harm an asset.

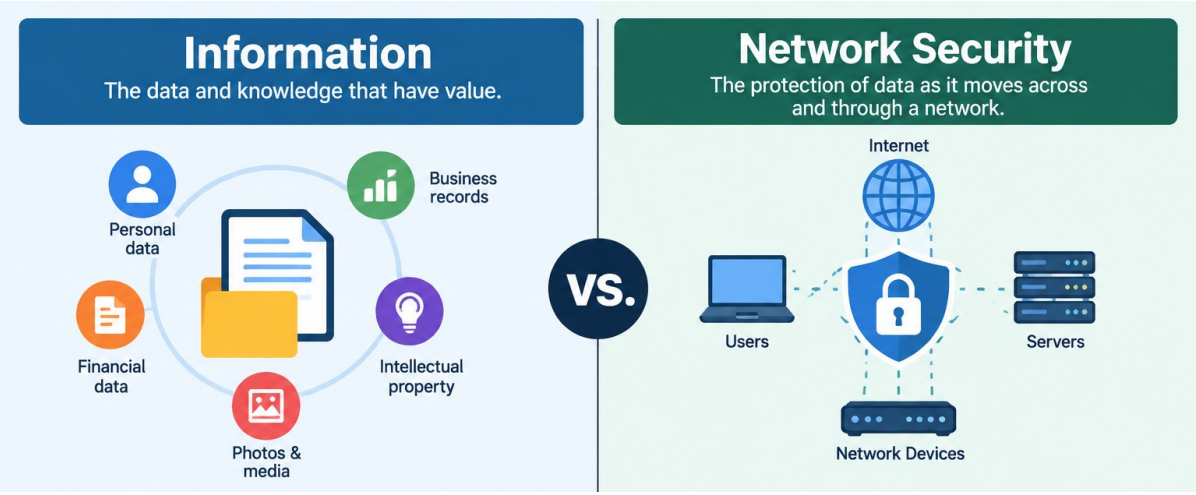


Source: *Diagram adapted from the risk model in NIST SP 800-30 Rev. 1, Guide for Conducting Risk Assessments*

Information vs. Network Security

There is no clean boundary between information security and network security — a threat often crosses both.

- **Example:** a virus can arrive physically on a USB drive or spread over the internet.
- **Either way,** once it infects a host, internal tools are needed to detect it and restore the system.
- **Lesson:** defenses must cover data at rest, data in transit, and the people who use them.



Layers to defend

- Physical media & devices
- Networks & transmission
- Applications & data
- People & processes

Security Services & Mechanisms

The OSI security architecture (ITU-T X.800) separates what we want to achieve from how we achieve it.

Security services — the goals

- Authentication — prove identity
- Access control — limit who uses what
- Data confidentiality — prevent disclosure
- Data integrity — detect changes
- Non-repudiation — no denying actions

Security mechanisms — the tools

- Encryption (encipherment)
- Digital signatures
- Access-control lists & policies
- Hashes & integrity checks
- Audit trails & monitoring

Who Are the Attackers?

Threat actors differ in skill, resources, and motive — and that shapes which defenses matter most.

Script kiddies	Cybercriminals	Hacktivists	Nation-states	Insiders
Low skill; use ready-made tools for fun or bragging rights.	Profit-driven: ransomware, fraud, stolen data sales.	Politically or socially motivated disruption and leaks.	Well-funded, patient espionage and sabotage (APTs).	Employees or partners who misuse legitimate access.

Thinking About Risk

Security decisions are risk decisions. Risk grows with the threat, the vulnerability it can exploit, and the impact if it succeeds.

- **Risk:** the chance that a threat exploits a vulnerability and causes harm.
- **Reduce it by:** removing vulnerabilities, adding controls, or limiting the impact.
- **Think both ways:** like an attacker who finds weaknesses and a defender who layers protection.

Risk factors

- Threat — who or what attacks
- Vulnerability — the weakness
- Impact — the damage done
- Likelihood — how probable

PART

02

The CIA Triad

20 minutes

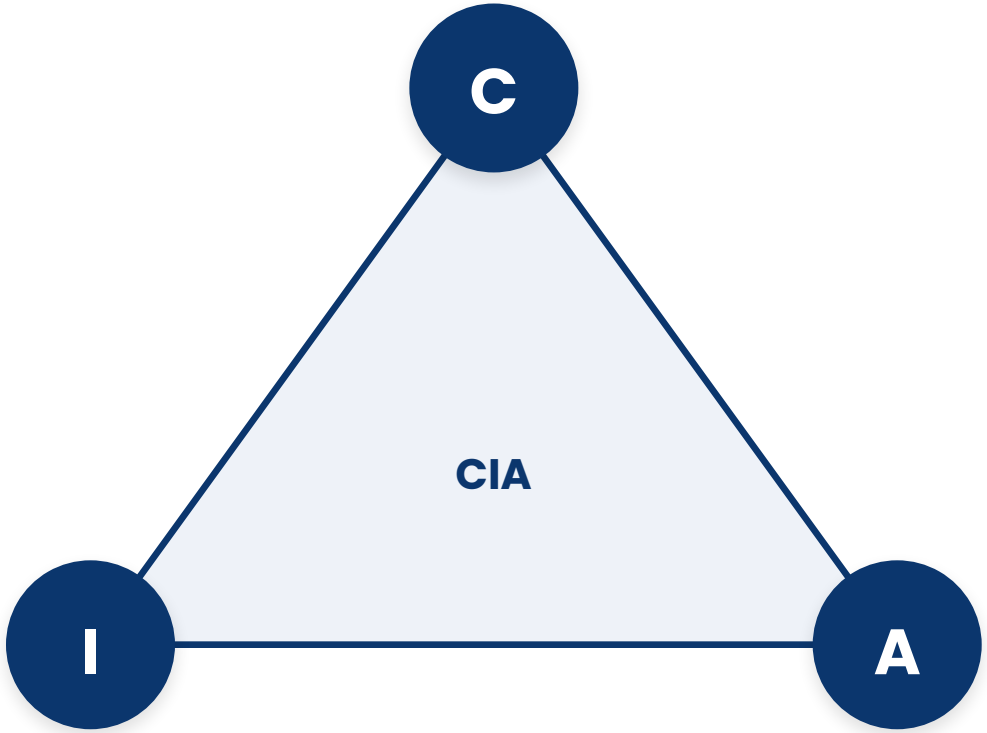
IN THIS PART

01 The CIA Triad
02 Confidentiality
03 Integrity
04 Availability
05 Beyond the CIA Triad
06 Worked Example: CIA at an ATM
07 Levels of Impact
08 Balancing the Triad
09 Mapping Attacks to the CIA Triad



The CIA Triad

The CIA triad is the lens used throughout the whole course. Most real incidents touch more than one of its three principles.



Confidentiality
Only authorized users can access data.

Integrity
Data stays accurate and unaltered.

Availability
Systems and data are usable when needed.

Confidentiality

Confidentiality ensures that information is disclosed only to those authorized to see it.

- **Data confidentiality:** private information is not made available to unauthorized individuals.
- **Privacy:** individuals control what data about them is collected and shared, and with whom.
- **Example:** a patient's medical records must be visible only to their care team.

Achieved with

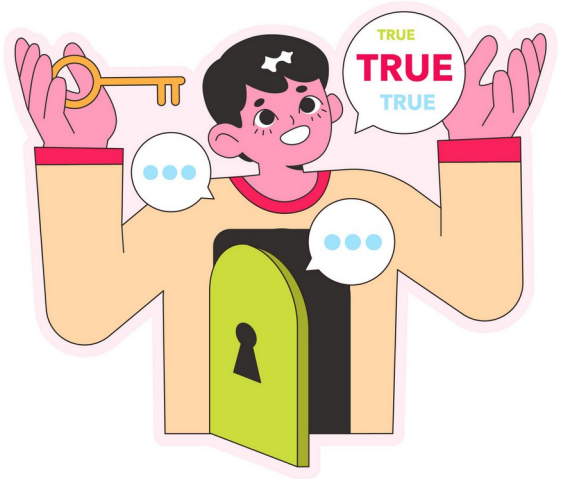
- Encryption
- Access control & authentication
- Data classification
- Need-to-know policies



Integrity

Integrity assures that information and systems are changed only in authorized ways.

- **Data integrity:** information and programs are changed only in a specified, authorized manner.
- **System integrity:** a system performs its function free from unauthorized manipulation.
- **Example:** an attacker altering a bank balance is an integrity violation.



Achieved with

- Hashing & checksums
- Digital signatures
- Version control & logging
- Change management

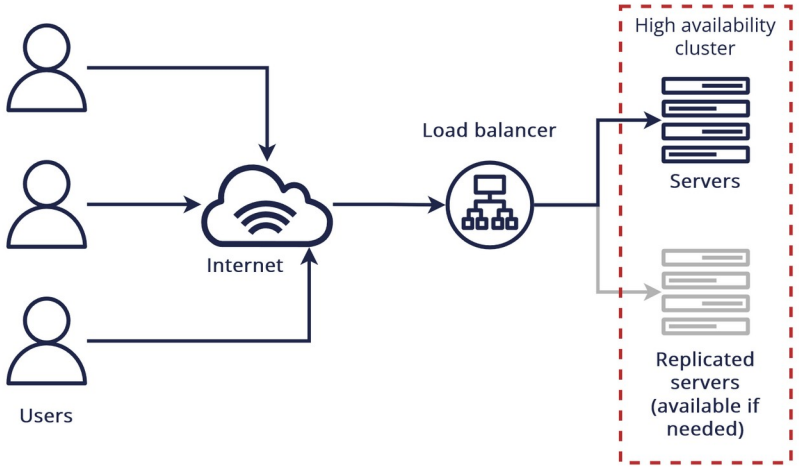
Availability

Availability ensures that systems and services operate reliably and are accessible to authorized users when needed.

- **Goal:** no disruption to legitimate access and timely, dependable service.
- **Example:** a ransomware attack that locks patient records is an availability failure.
- **Threats:** denial-of-service attacks, hardware failure, and disasters.

Achieved with

- Redundancy & failover
- Backups & recovery plans
- Load balancing
- DoS protection



Beyond the CIA Triad

Many experts add two further concepts for a fuller picture of security.

Authenticity

Assurance that something is genuine and can be verified that users and inputs are who and what they claim to be.

Accountability

Every action can be traced to a responsible entity, supporting nonrepudiation, auditing, and incident investigation.

Worked Example: CIA at an ATM

Consider an ATM where a user inserts a card and enters a PIN. Each CIA principle maps to a concrete requirement and a level of importance.

Confidentiality

The PIN and account details must never be revealed to others.

Importance: HIGH



Integrity

Transactions and balances must be recorded and updated exactly.

Importance: HIGH



Availability

The machine should be usable whenever customers need cash.

Importance: MODERATE



Levels of Impact

A breach is judged by how badly it would affect the organization — this guides how much protection each asset needs.

Low	Moderate	High
Minor disruption and small losses; the organization keeps functioning.	Notable loss of performance, finances, or reputation — but not catastrophic.	Severe disruption: major loss, serious harm, or a complete shutdown.

Balancing the Triad

The three principles can pull against each other. Good security finds the right balance for each system.

- **Confidentiality vs. availability:** stricter access checks and encryption can slow or block legitimate users.
- **Integrity vs. availability:** extra validation and approvals add delay before changes go live.
- **Security vs. usability:** rules that are too painful get bypassed, which weakens all three.
- **Decide by impact:** a hospital favors availability; a bank ledger favors integrity.

Questions to ask

- What matters most here?
- What is the cost of downtime?
- What is the cost of a leak?
- Will users work around it?

Mapping Attacks to the CIA Triad

Every attack ultimately targets one or more CIA principles. Naming which one helps you pick the right defense.

Attack	Principle at risk	Typical defense
Eavesdropping / sniffing	Confidentiality	Encryption (TLS, VPN)
Tampering / forgery	Integrity	Hashing, digital signatures
Denial of service	Availability	Redundancy, filtering
Impersonation / spoofing	Authenticity	Strong authentication
Repudiation of an action	Accountability	Audit logs, non-repudiation

PART

03

Common Network Threats

20 minutes

IN THIS PART

01 The Threat Landscape
02 Passive Attacks
03 Active Attacks
04 Malware
05 Phishing & Social Engineering
06 Man-in-the-Middle & Spoofing
07 Denial of Service (DoS)
08 Insider Threats
09 Types of Network Security Attack
10 Defense in Depth



The Threat Landscape

Attacks are commonly classified by how they interact with the data stream and real breach data shows how attackers actually get in.

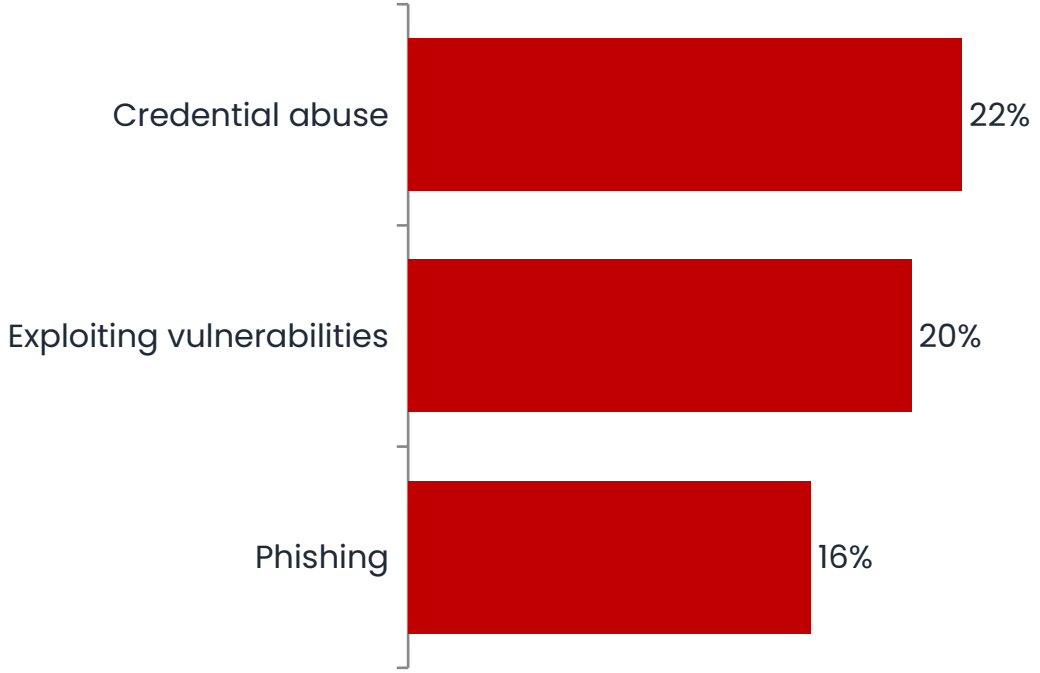
Passive attacks

Eavesdrop or monitor without altering data — e.g., intercepting messages or analyzing traffic. Hard to detect; prevented with encryption.

Active attacks

Modify the data stream or create false traffic — masquerade, replay, message modification, denial of service.

Top initial access vectors in breaches, 2025

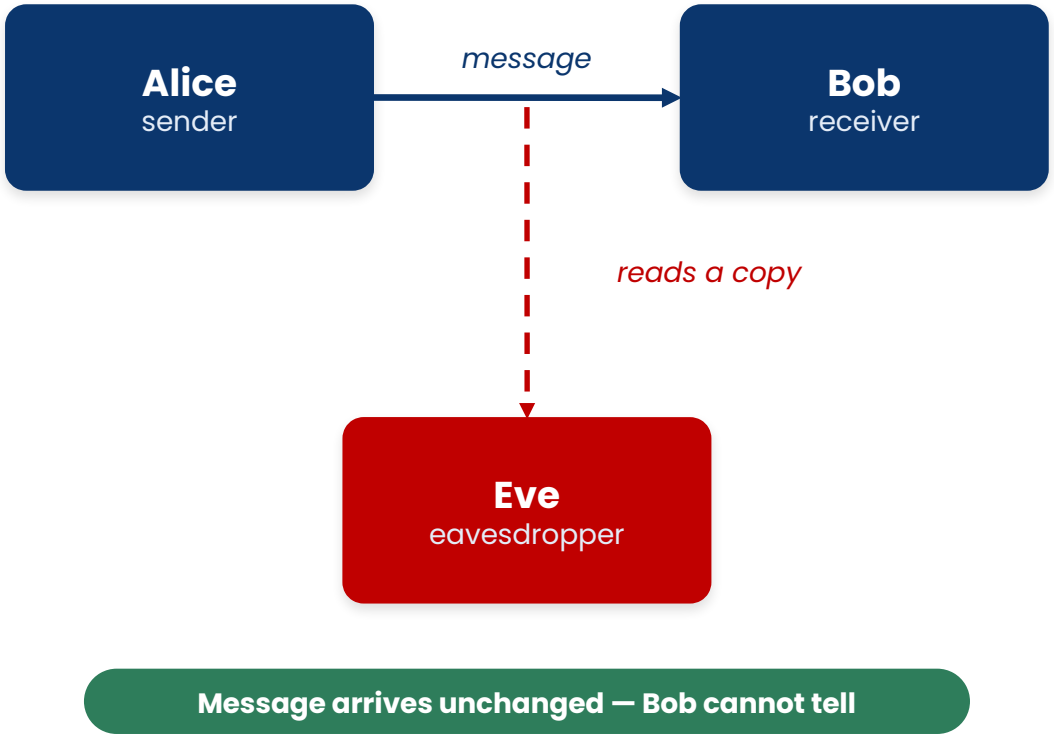


Source: *Verizon, 2025 Data Breach Investigations Report (12,195 confirmed breaches)*

Passive Attacks

Passive attacks observe traffic without altering it. They are hard to detect, so the emphasis is on prevention.

- **Release of message contents:** intercepting sensitive communications such as emails or calls.
- **Traffic analysis:** inferring information from patterns who talks to whom, when, and how much / even when the content is encrypted.
- **Prevention:** strong encryption (TLS, VPN), traffic padding, and minimizing exposed metadata.



Source: *Adapted from ITU-T Rec. X.800 and W. Stallings, Cryptography and Network Security*

Active Attacks

Active attacks modify the data stream or inject false traffic. They are easier to detect but harder to prevent outright.



Masquerade

One entity pretends to be another to gain access.

Replay

Capturing valid data and resending it for an unauthorized effect.

Modification

Altering or reordering messages to change their meaning.

Denial of Service

Disrupting or blocking legitimate use of a service.

Source: *Adapted from ITU-T Rec. X.800 and W. Stallings, Cryptography and Network Security*

Malware

Malware is malicious software designed to damage, disrupt, or gain unauthorized access.



Virus	Worm	Trojan	Ransomware	Spyware
Attaches to a file and spreads when it runs.	Self-replicates across a network on its own.	Hides malicious code inside useful-looking software.	Encrypts data and demands payment to release it.	Secretly collects information about the user.

Phishing & Social Engineering

Social-engineering attacks target people rather than technology, tricking them into breaking security.

- **Phishing:** fraudulent emails or sites that trick users into revealing credentials.
- **Spear phishing:** a targeted version aimed at a specific person or role.
- **Pretexting / baiting:** inventing a scenario or lure to obtain information or access.



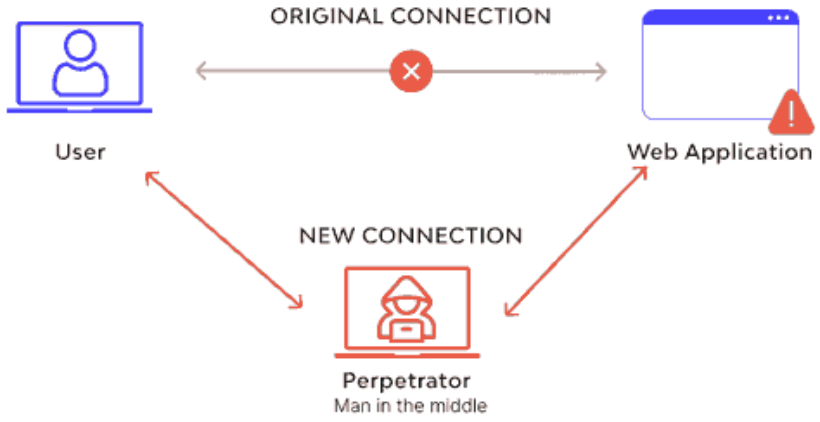
Defenses

- User awareness training
- Email filtering
- Multi-factor authentication
- Verify before you trust

Man-in-the-Middle & Spoofing

These attacks place the attacker between two parties, or forge a trusted identity to slip past defenses.

- **Man-in-the-middle:** the attacker secretly relays and can alter traffic between two parties who believe they communicate directly.
- **Spoofing:** forging an address or identity (IP, email, ARP, or DNS) to appear trusted.
- **Defense:** authentication, encryption (TLS), and integrity checks.



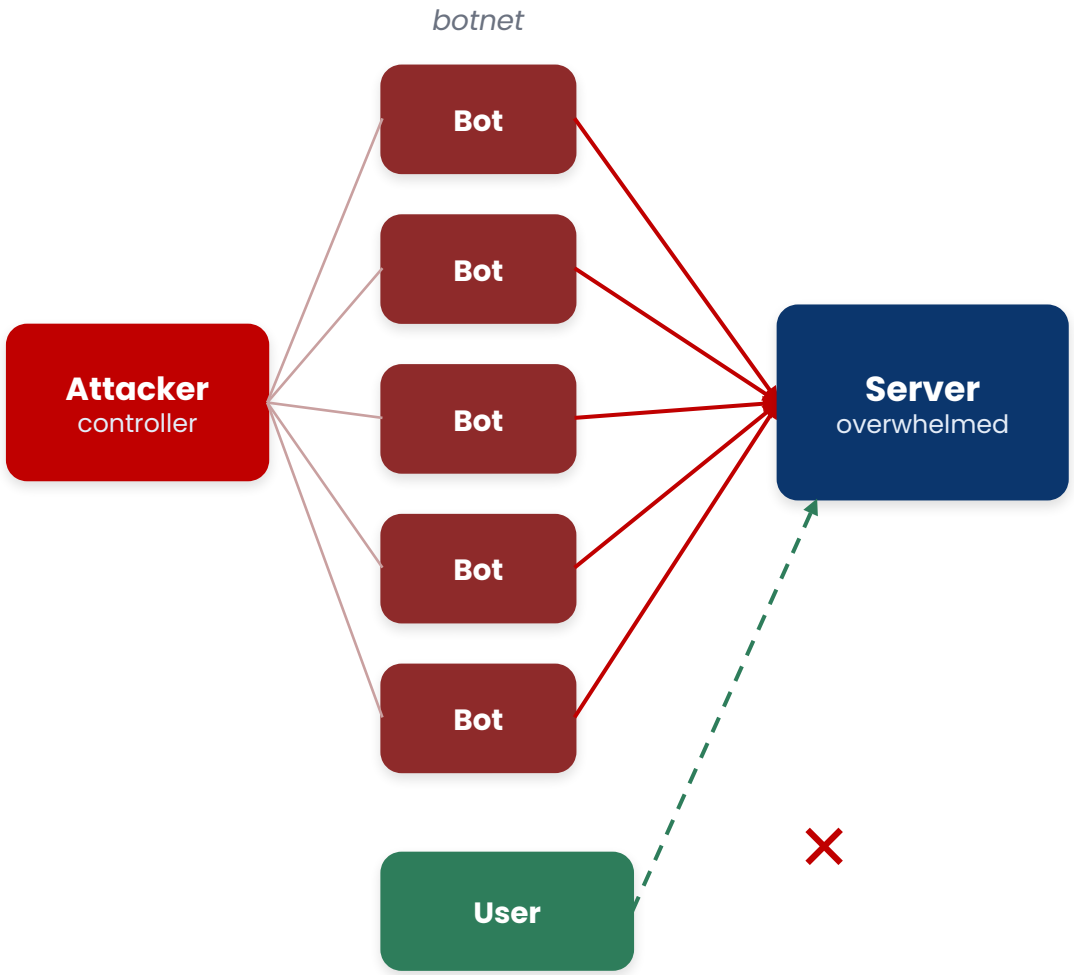
Where it happens

- Public / open Wi-Fi
- Unencrypted sessions
- ARP or DNS poisoning
- Forged email headers

Denial of Service (DoS)

A denial-of-service attack overwhelms a network or service so legitimate users cannot reach it — a direct attack on availability.

- **DoS:** a single source floods the target with traffic or requests.
- **DDoS:** many compromised machines (a botnet) attack at once, amplifying the effect.
- **Impact:** downtime, lost revenue, and cover for other attacks.
- **Defenses:** rate limiting and filtering, traffic-scrubbing services, CDNs and over-provisioning, an incident response plan.



Source: *Adapted from CISA, Understanding Denial-of-Service Attacks*

Insider Threats

Not every threat is external — insiders already hold authorized access, which makes their misuse hard to detect.

- **Malicious insiders:** employees who deliberately steal or sabotage data.
- **Negligent insiders:** well-meaning users who make careless mistakes.
- **Compromised accounts:** legitimate credentials stolen and misused by an attacker.

Defenses

- Least-privilege access
- Monitoring & audit logs
- Separation of duties
- Off-boarding controls



Types of Network Security Attack

Putting it together — real-world attacks fall into four broad families:

Malware

- Virus
- Worm
- Trojan
- Ransomware

Social Engineering

- Phishing
- Pretexting
- Baiting

Network Attacks

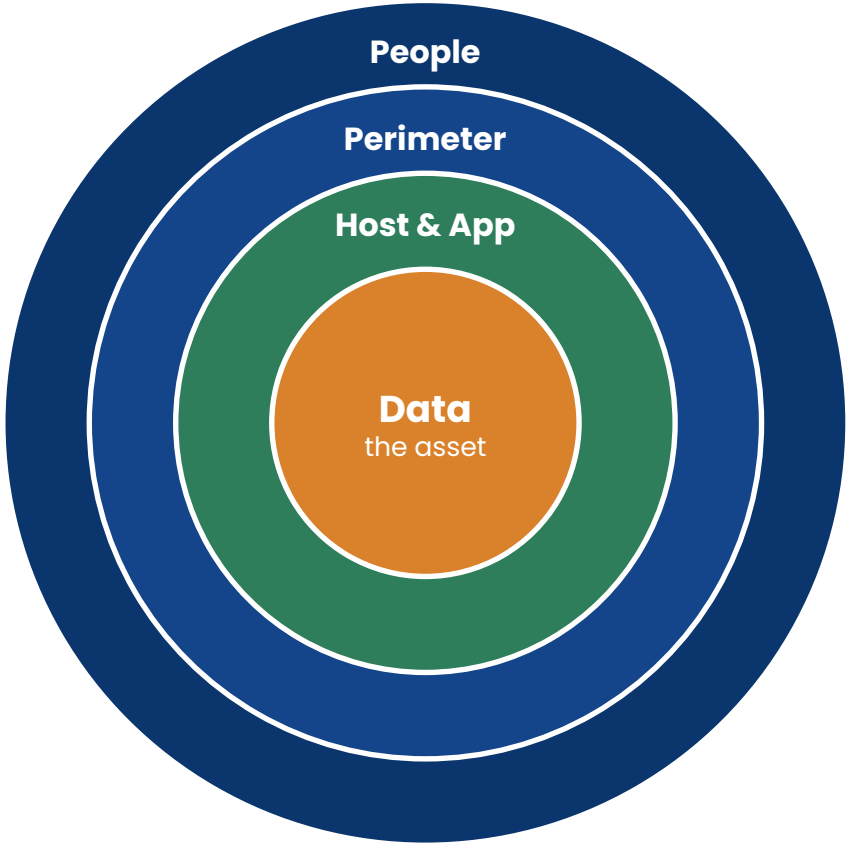
- DoS / DDoS
- Man-in-the-Middle
- Spoofing

Web & Access

- SQL Injection
- Brute Force
- Privilege Escalation

Defense in Depth

No single control stops every attack. Layered defenses mean that when one layer fails, others still protect the asset.



People
Training, policies, and awareness.

Perimeter
Firewalls, IDS/IPS, and segmentation.

Host & App
Patching, hardening, and access control.

Data
Encryption, backups, and monitoring.

Source: *Concept per NIST CSRC Glossary, "defense-in-depth"*

Case Study: The Equifax Breach (2017)

147M

people had personal data exposed

Attackers exploited an unpatched web-application vulnerability.

Confidentiality failed
Private records were exposed to attackers.

Preventable
A known patch existed but was not applied in time.

Discussion: Which part of the CIA triad failed in this case, and what preventive measures could have helped?

Summary

- ✓ Network security protects systems and data from a range of attacks.
- ✓ The CIA triad forms the foundation of all security principles.
- ✓ Real-world breaches highlight the importance of proactive, layered defense.

Post-Quiz / Check Understanding

Discuss each question before revealing the answer.

1. Which principle ensures data is accessible when needed?

a) Authentication
b) Confidentiality
c) Availability
d) Integrity

2. Keeping data accurate and unaltered is:

a) Redundancy
b) Confidentiality
c) Integrity
d) Availability

3. A DoS attack primarily targets:

a) Non-repudiation
b) Confidentiality
c) Integrity
d) Availability

4. The Equifax breach was caused mainly by:

a) A phishing email to the CEO
b) A weak Wi-Fi password
c) An unpatched software vulnerability
d) A stolen laptop

5. Studying past breaches helps to:

a) Replace firewalls
b) Improve security awareness and prevention
c) Slow down networks
d) Avoid encryption

Post-Quiz — Answers

Correct answers are highlighted.

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References & Next Lesson

REFERENCES

- Stallings, W. Network Security Essentials: Applications and Standards — Ch. 1 (Computer Security Concepts, OSI Security Architecture).
- U.S. GAO (2018). Equifax Data Breach report.
- NIST SP 800-12 Rev. 1: An Introduction to Information Security.

NEXT LESSON

In the next session you will study Cryptography Basics and explore how encryption helps maintain confidentiality in network communications.

Figure & Data Sources

All diagrams were drawn for this course. Charts use the published figures cited below.

Slide	Figure	Source
10	Average cost of a data breach, 2020–2025	IBM Security, Cost of a Data Breach Report (2020–2025 editions) www.ibm.com/reports/data-breach
12	Threat–vulnerability–asset–control relationship	Diagram adapted from the risk model in NIST SP 800-30 Rev. 1, Guide for Conducting Risk Assessments csrc.nist.gov/pubs/sp/800/30/r1/final
28	Top initial access vectors in breaches	Verizon, 2025 Data Breach Investigations Report (12,195 confirmed breaches) www.verizon.com/business/resources/reports/dbir/
29	Passive attack (eavesdropping)	Adapted from ITU-T Rec. X.800 and W. Stallings, Cryptography and Network Security www.itu.int/rec/T-REC-X.800-199103-I
30	Active attack (interception and modification)	Adapted from ITU-T Rec. X.800 and W. Stallings, Cryptography and Network Security www.itu.int/rec/T-REC-X.800-199103-I
34	Distributed denial of service (botnet)	Adapted from CISA, Understanding Denial-of-Service Attacks www.cisa.gov/news-events/news/understanding-denial-service-attacks
37	Defense-in-depth layers	Concept per NIST CSRC Glossary, “defense-in-depth” csrc.nist.gov/glossary/term/defense_in_depth

Thank you!

